

1 Solution — GIAM 3.2

Problem 6

1.1 Problem

Prove that for all real numbers a, b, c : if $ac < 0$, then the quadratic equation $ax^2 + bx + c = 0$ has two real solutions.

Hint. The equation $ax^2 + bx + c = 0$ has two real solutions if and only if $b^2 - 4ac > 0$ and $a \neq 0$.

1.2 Proof

***Proof.** Suppose $ac < 0$. By the hint there are two things to establish.*

***First,** $a \neq 0$. Suppose instead that $a = 0$. Then $ac = 0 \cdot c = 0$, which is not less than 0 — contradicting the hypothesis. Hence $a \neq 0$.*

***Second,** $b^2 - 4ac > 0$. Since $ac < 0$ and -4 is negative, multiplying reverses the inequality:*

$$-4ac > 0.$$

Also b is real, so $b^2 \geq 0$. Adding,

$$b^2 - 4ac = \underbrace{b^2}_{\geq 0} + \underbrace{(-4ac)}_{> 0} > 0.$$

Both clauses of the criterion hold, so the equation has two real solutions. ■

1.3 Two Clauses, Two Obligations

The criterion has two clauses, so the proof has two obligations. The single hypothesis $ac < 0$ supplies both.

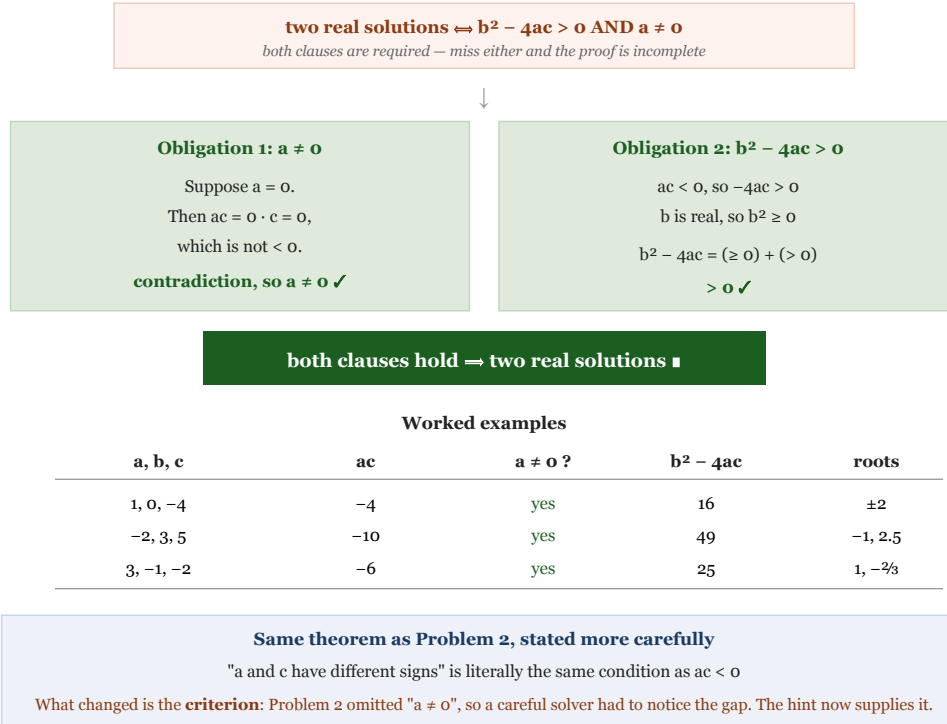


Figure 1.1: Top: the criterion — two real solutions if and only if b squared minus $4ac$ is positive AND a is nonzero — noting both clauses are required and missing either leaves the proof incomplete. Two green panels below. Obligation 1, a nonzero: suppose a were 0; then ac equals 0 times c equals 0, which is not less than 0, a contradiction, so a is nonzero. Obligation 2, b squared minus $4ac$ positive: ac is negative so minus $4ac$ is positive; b is real so b squared is at least zero; hence b squared minus $4ac$ is a non-negative plus a positive, which is positive. A green banner: both clauses hold, so there are two real solutions. A table of worked examples: a, b, c equal 1, 0, minus 4 gives ac equal minus 4, a nonzero yes, discriminant 16, roots plus and minus 2; minus 2, 3, 5 gives ac minus 10, discriminant 49, roots minus 1 and 2.5; and 3, minus 1, minus 2 gives ac minus 6, discriminant 25, roots 1 and minus two thirds. A closing panel: this is the same theorem as Problem 2 stated more carefully — ‘ a and c have different signs’ is literally the same condition as ac less than zero, and what changed is the criterion, since Problem 2 omitted the a nonzero clause so a careful solver had to notice the gap, whereas the hint now supplies it.

The structure worth noticing is that the criterion is a **conjunction**, so proving it requires proving *both* conjuncts. A proof that establishes only $b^2 - 4ac > 0$ is incomplete, however convincing it looks.

1.4 Remark — This Is Problem 2, Stated Correctly

The hypothesis here, $ac < 0$, is *literally the same condition* as Problem 2's " a and c have different signs" — a product of two reals is negative exactly when the factors have opposite signs (checked over $[-20, 20]^2$). So the two problems state the same theorem.

What changed is the **criterion**. Problem 2 quoted it as “two real solutions iff $b^2 - 4ac$ is positive,” omitting $a \neq 0$. That version is strictly false: $0x^2 + 1x + (-1) = 0$ has discriminant $1 > 0$ but is the linear equation $x = 1$, with **one** solution, not two.

Problem 2's solution flagged this in passing; here the hint promotes it to a required step. That is the pedagogical point of restating the problem — the same mathematics, but now you cannot get full credit without noticing the second clause.

1.5 Remark — Why $a \neq 0$ Comes Free

The first obligation is discharged in one line because “having a sign” and “being nonzero” are the same thing. Formally: $ac < 0$ implies $ac \neq 0$, and a product of reals is nonzero only if both factors are — so $a \neq 0$ (and $c \neq 0$ too, though that is not needed).

This is the “no zero divisors” property that has recurred throughout — in Problem 3.1.3 it guaranteed $bd \neq 0$, and in Problem 3.1.13 it licensed cancellation. Here it is used in its contrapositive form: a nonzero product forces nonzero factors.

1.6 Remark — The Argument Is Insensitive to b

Notice b appears in the proof exactly once, contributing only the inequality $b^2 \geq 0$. Nothing about its size, sign, or whether it is zero matters.

That is unusual and worth appreciating: the theorem holds for *every* real b simultaneously. Geometrically (as Problem 2 discussed) b merely slides the parabola sideways, changing neither the opening direction nor the value at $x = 0$ — and those are the only two features the argument uses. The case $b = 0$ is included and unremarkable: $x^2 - 4 = 0$ has $ac = -4 < 0$, discriminant 16 , and roots ± 2 .

1.7 Remark — What a Two-Part Criterion Demands of a Disproof

Since the criterion is a conjunction, its **negation** is a disjunction (Problem 2.3.5):

$$\neg(b^2 - 4ac > 0 \wedge a \neq 0) \cong (b^2 - 4ac \leq 0) \vee (a = 0).$$

So an equation fails to have two real solutions if *either* the discriminant is non-positive *or* it is not really quadratic. Both failure modes occur:

example	which clause fails
$x^2 + 1 = 0$	discriminant $= -4 \leq 0$
$x^2 - 2x + 1 = 0$	discriminant $= 0$ — a repeated root, so <i>one</i> solution
$0x^2 + x - 1 = 0$	$a = 0$ — linear, one solution

Table 1.1

Keeping the two clauses distinct is what lets you say *which* thing went wrong.

1.8 Remark — The Converse Still Fails

As in Problem 2, the implication runs one way only. Take $a = 1$, $b = 5$, $c = 1$: the discriminant is $25 - 4 = 21 > 0$ and $a \neq 0$, so there are two real solutions — yet $ac = 1 > 0$.

The hypothesis $ac < 0$ is a **convenient sufficient condition**: it makes both terms of $b^2 - 4ac$ non-negative, so no cancellation can occur and the discriminant is positive regardless of b . When $ac > 0$, the term $-4ac$ opposes b^2 and the outcome depends on which wins. That is exactly the trade the hypothesis is designed to avoid, and it is why the theorem can dispense with any assumption on b .